DES Pune University, Pune

School of Engineering and Technology
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Assignment No.:	7	
Title:	**To write a C++ program to add a friend function for more than one class.**	
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Contents

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Theory (Logic)

This is a C++ program shows how to use a friend. The cat class has private (priv) and protected (pro) variables. Normally, other classes can't access them. But by using the friend keyword, the cat class allows the animal class to access its hidden data. The animal class has a function display that prints those values. In main, we create objects of both classes and call the display function to show the values.

Pseudo Code

- **Step 1: Include <iostream>**
- **Step 2: Use the standard namespace**
- **Step 3: Create Class Cat**
    - **Private:**
    - **Integer priv**
    - **Protected:**
    - **Integer pro**
    - **Public:**
    - **Constructor:**
    - **Set priv = 10**
    - **Set pro = 20**
- **Step 4: Declare Animal as a friend class**
- **Step 5: Class Animal:**
    - **Public:**
    - **Function display(CatObject):**
    - **Print value of CatObject.priv**
    - **Print value of CatObject.pro**
- **Step 6: Start Main Function**
    - **Create object of Cat**
    - **Create object of Animal**
    - **Call Animal's display function and pass Cat object**
- **Step 7: End main function**

Program Code

```
#include <iostream>
using namespace std;

class cat {
private:
   int priv;

protected:
   int pro;

public:
   cat() {
      priv = 10;
      pro = 20;
   }
   friend class animal;
};

class animal {
public:
   void display(cat &t) {
      cout << "The value of Private Variable = "<< t.priv << endl;
      cout << "The value of Protected Variable = "<< t.pro;
   }
};

int main() {
   cat obj;
   animal obj1;
   obj1.display(obj);
   return 0;
}
```

Output

```
The value of Private Variable = 10
The value of Protected Variable = 20
```

Conclusion

Code is run successfully, required output is received.